IGNORE:

🔹 **Create folder:**

mkdir IGNORE

🔹 **Go inside folder:**

cd IGNORE

🔹 **Create files:**

touch test1.txt
touch test2.txt

👉 Go back to main folder:

cd ..

✅ **STEP g: Create .gitignore File**

touch .gitignore

👉 Open .gitignore and write:

IGNORE/

✓ This tells Git to **ignore the entire IGNORE folder**

❗ **IMPORTANT (Very Important Step)**

👉 If files were already added before .gitignore, remove them from tracking:

git rm -r --cached IGNORE

✅ **STEP h: Commit & Push**

git add .
git commit -m "Added .gitignore to ignore IGNORE folder"
git push

DESKTOP:

📁 **STEP 3: Create New Repository**

1. Click **File → New Repository**
2. Fill details:
    - Name: DataScienceProject
    - Local Path: choose any folder (e.g. Desktop)
3. Click **Create Repository**

✔ Now your project is ready locally

☁️ **STEP 4: Publish to GitHub**

1. Click **Publish repository** (top right)
2. Choose Public/Private
3. Click **Publish Repository**

✔ Now your repo is online

🌿 **STEP 5: Create Branch (GIT-EXAM-1)**

1. Click **Current Branch**
2. Click **New Branch**
3. Name: GIT-EXAM-1
4. Click **Create Branch**

💻 **STEP 6: Add Python Programs**

👉 **Open project folder:**

Click **Show in Explorer**

🔹 **Create program2.py (Palindrome)**

```
string = input("Enter string: ")

if string == string[::-1]:
    print("Palindrome")
else:
    print("Not Palindrome")
```

🔹 **Create program1.py (Squares)**

```
for i in range(1, 11):
    print(i*i)
```

💾 STEP 7: Commit & Push

1. Go back to GitHub Desktop
2. You will see both files listed
3. Write commit message:

Added program1 and program2

4. Click **Commit to GIT-EXAM-1**
5. Click **Push origin**

📂 STEP 8: Create IGNORE Folder

1. Click **Show in Explorer**
2. Create folder: IGNORE
3. Inside it create:
    - test1.txt
    - test2.txt

🚫 STEP 9: Create .gitignore

1. In same folder create file: .gitignore
2. Add this line:

IGNORE/

🔄 STEP 10: Commit & Push Again

1. Go to GitHub Desktop
2. Only .gitignore will appear (not IGNORE folder)
3. Commit message:

Added gitignore file

4. Click **Commit**
5. Click **Push origin**

🔍 **STEP 11: Verify on GitHub**

Open your repo in browser

You will see:

- ✅ program1.py
- ✅ program2.py
- ✅ .gitignore

CLONE:

✅ **a. Clone Repository**

1. Open **GitHub Desktop**
2. Click **File → Clone Repository**
3. Go to **URL tab**
4. Paste:

https://github.com/KalyanM45/AI-Project-Gallery.git

5. Choose local path → Click **Clone**

❌ **b. Demonstrate Branch Creation is Disallowed**

1. In GitHub Desktop → Click **Current Branch**
2. Click **New Branch**
3. Name: test-branch
4. Click **Create Branch**

👉 Now click **Publish Branch**

❌ You will get error like:

- "Permission denied"
- "Cannot push to this repository"

✅ **Conclusion:** You don't have write access → branch creation is restricted

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🔱 c. Fork Repository

1. Open the repo page in browser
2. Click **Fork**
3. It creates your copy (your account repo)

📥 Clone Your Fork (Important)

1. In GitHub Desktop → **File → Clone Repository**
2. Go to **GitHub.com tab**
3. Select your forked repo
4. Click **Clone**

🌿 d. Create Feature Branch (GIT-EXAM)

1. Click **Current Branch**
2. Click **New Branch**
3. Name: GIT-EXAM
4. Click **Create Branch**

✏️ e. Add / Modify Code

1. Click **Show in Explorer**
2. Create file:

program_exam.py

3. Add code:

```
print("Hello from GIT-EXAM branch")
```

4. Save file

💾 Commit Changes

Back to GitHub Desktop:

1. You will see changes automatically
2. Add message:

Added program_exam file

3. Click **Commit to GIT-EXAM**

🚀 **Push Branch**

Click:

👉 **Publish Branch**

🔁 **f. Create Pull Request**

1. Click **Create Pull Request**
2. Browser opens
3. Add title & description
4. Click **Create Pull Request**

⚠️ **g. Resolve Merge Conflicts**

If conflict occurs:

Step 1: Update branch

- Click **Fetch origin**
- Click **Pull origin**

Step 2: If conflict shown

- GitHub Desktop shows conflicted files
- Click **Open in Editor**

Step 3: Fix conflict manually

```
<<<<<<< HEAD
your code
=======
other code
>>>>>>> main
```

👉 Remove markers and keep correct code

Step 4: Mark resolved

1. Back to GitHub Desktop
2. Click **Mark as Resolved**
3. Click **Commit Merge**

Step 5: Push again

👉 Click **Push origin**

✅ **Final Exam Summary (Short)**

- Clone repo using GitHub Desktop
- Branch creation fails due to permission
- Fork repository
- Clone fork
- Create branch (GIT-EXAM)
- Add/modify code
- Commit & publish branch
- Create pull request
- Resolve conflicts and push updates